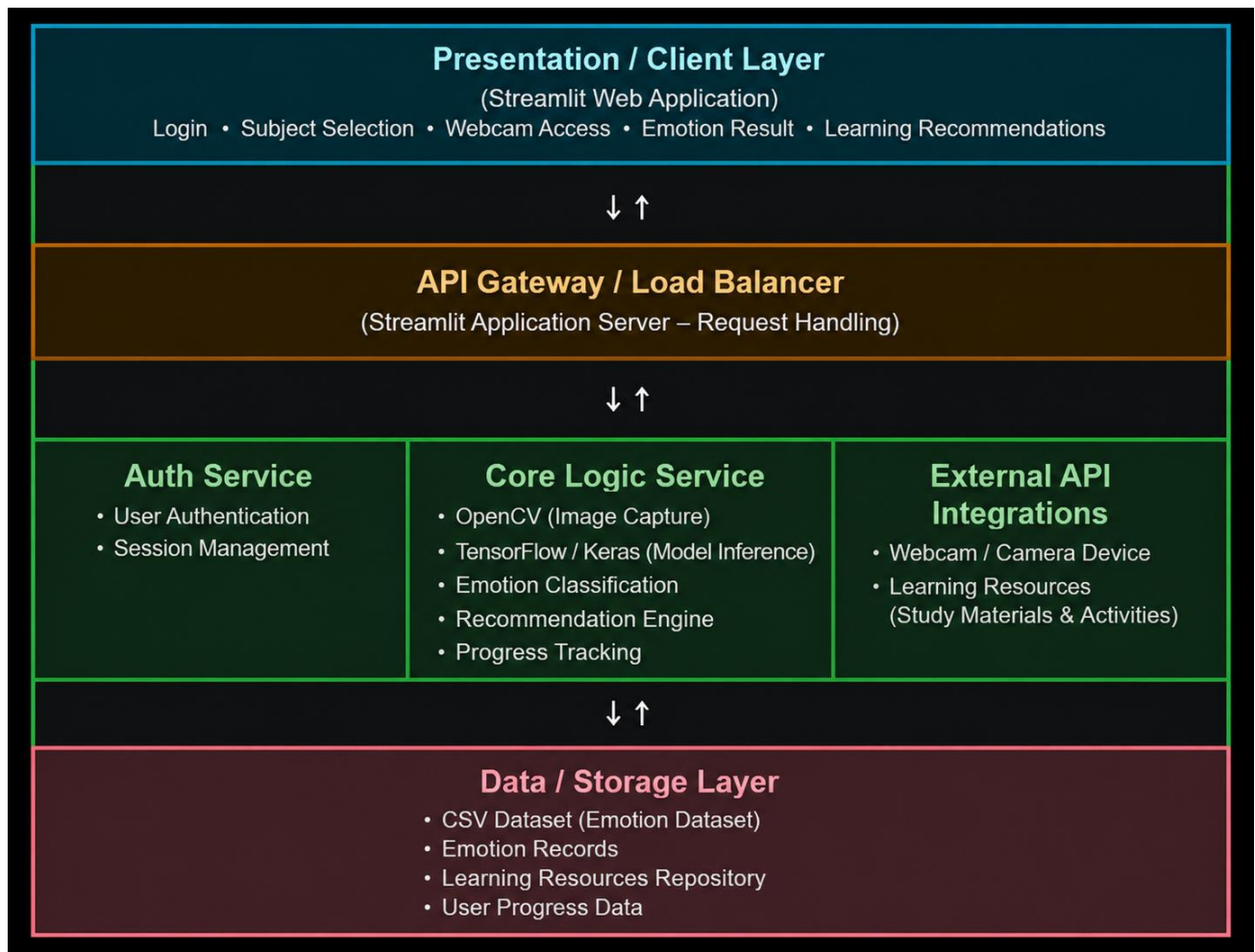


Solution Architecture

Date	06 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface for emotion detection, subject selection, and personalized learning recommendations.</i>	<i>Streamlit</i>
<i>API Gateway</i>	<i>Handles user requests and routes them to the application logic.</i>	<i>Streamlit Server</i>
<i>Microservices</i>	<i>Detects emotions using AI, processes facial images, and generates personalized learning recommendations.</i>	<i>Python, OpenCV, TensorFlow/Keras, NumPy, Pandas</i>
<i>Database</i>	<i>Stores datasets, detected emotions, learning resources, and user progress.</i>	<i>CSV Files</i>